

Yiming Xu

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English CET-4 / CET-6

Profile

Dalian University of Technology
B.Eng. in Software Engineering, 2023.09 – Present
Research interests: **multimodal learning, agent systems**, and computer vision
GPA: **85.8** Rank: **20/120**

Awards

19th National Undergraduate Software Innovation Competition
Second Prize, Northeast Region 2026

Chinese Collegiate Computing Competition (4C), Liaoning
Multiple Second / Third Prizes 2026

Chinese Collegiate Computing Competition (4C)
Campus Second Prize 2026

Mathematical Contest in Modeling (MCM/ICM)
Meritorious Winner 2025

China Collegiate Digital Media Technology Competition
National Third Prize 2025

Asia-Pacific Mathematical Contest in Modeling
Second Prize 2025

"CSEE Cup" National Electrical Engineering Math Modeling Competition
Second Prize 2024

Asia-Pacific Mathematical Contest in Modeling
Third Prize 2024

Internship Experience

Meituan - Tax and Finance R&D Intern 2026.01 – 2026.03

- Built a tax-domain **RAG agent** with LangChain, vector retrieval, and document chunking, achieving **85%+** retrieval accuracy.
- Contributed to MCP + Skills SOP automation, implementing tool-call workflows for login, page recognition, and form filling.
- Processed and organized **10,000+** tax and regulatory documents, iterating retrieval pipelines to improve knowledge coverage and answer quality.
- Contributed to task-oriented agent workflow design and optimization, focusing on task decomposition, tool orchestration, and execution.

Libernovo - Large Language Model Intern 2026.10 – 2026.11

- Improved **Embodied Agent** tool use and task execution for **dual robotic arms, smart desks, and smart chairs**, under constraints on small-model capabilities and context length.
- Designed an **Agent–Tools execution pipeline** connecting task decomposition, tool orchestration, and real-device execution across multiple device types, and refined experiments using execution feedback.
- Collected **real-device data and analyzed failure causes**, converting execution traces and findings into **6,804 training examples** through a collection, analysis, and dataset-building loop.
- Used **SFT, DPO, and RL** for small-model fine-tuning and policy optimization, connecting data collection to iterative model training and improving task execution accuracy from **18.6% to 78.4%** (**+59.8 percentage points**).

Research Experience

GS-UVCE: Underwater Visual Enhancement and 3D Reconstruction 2026

IEEE ICRA 2026 (CCF-B) Accepted

- Built an end-to-end **3D Gaussian Splatting** framework for color shifts, low-light degradation, and multi-view inconsistency caused by underwater scattering, light attenuation, and non-uniform illumination. Jointly modeled the water medium, view-dependent lighting, and scene depth for unsupervised visual enhancement and high-fidelity 3D reconstruction.
- Contributed to **Light-MLP and enhancement objective design**, modeling view-dependent lighting as a ray-direction function and using dynamic Gamma correction to improve object colors, cross-view brightness and color consistency, and low-light detail recovery.
- Contributed to exposure, backscatter, and depth loss design; **designed and analyzed 9 ablation experiments** on Light-MLP, Medium-MLP, and depth regularization to quantify their effects on enhancement and reconstruction quality.
- Achieved approximately **9.6–12.9 dB PSNR gains** over vanilla 3DGS on four SeaThru-NeRF underwater scenes, with rendering speeds of **190.3 FPS**.

Skills

Programming	Python, C++, PyTorch
Multimodal	Prompt tuning, I2V, T2V
Agents	LangChain, RAG, vector retrieval, tool calling, Skills
Vision	3D vision, Gaussian Splatting, image enhancement
Languages	English: CET-4 / CET-6

Campus Leadership

President, Student Union of School of Software
Standing Representative, University Student Congress
President, Sunshine Mental Health Association
Representative, Youth League Congress

Research Experience (continued)

AdapI2V: Adaptive Multi-Agent Iterative Optimization for Image-to-Video Generation

2026

The Chinese University of Hong Kong, Shenzhen Excellent Undergraduate Summer Camp | *Target submission: CVPR 2027*

- Addressed **missing actions, incorrect action order, and cross-segment discontinuities** in image-to-video generation under complex instructions, improving multi-action completion and video coherence **without training the underlying models and within a limited generation budget**.
- Designed a collaborative framework of **planning, verification, and repair agents**. Parsed instructions into action-event graphs, **adaptively determined segment counts and stage goals**, and passed final-frame states to subsequent generations to preserve action order and continuity.
- Built **structured evaluation with 28 criteria across seven dimensions and targeted prompt repair** to locate action, state, and continuity failures. Allocated retry budgets within **at most 5 generation calls per valid execution**, with fallback to original candidates to safeguard overall video quality.
- On **650 cases** from the adapted TIP300 / WMB350 evaluation sets, improved Global28QA scores over VPO + BoN5 by **15.45 / 5.22 points**, respectively. Reduced generated frame counts for valid executions by **49.7% / 53.7%** relative to five-candidate sampling.

Project Experience

Simulation-Driven Multimodal Data Collection and Processing Platform

2025–2026

National Undergraduate Innovation and Entrepreneurship Training Program | 1 Software Copyright

- Built a **CARLA, Gazebo, and ROS2** platform to unify multi-sensor data collection, management, and reuse in simulation, providing replayable and verifiable data for downstream perception model training.
- Developed Python tools for **ROS2 Bag parsing, sensor data organization, format conversion, and basic quality checks**, creating a workflow from simulation data collection to preprocessing.
- Completed **CARLA–ROS2 interface integration, recording, and replay validation**; contributed to troubleshooting and workflow standardization, supporting a software copyright registration for the tools.

Honors

2024–2025: **Social Work Scholarship, Technological Innovation Scholarship, Social Practice Scholarship, Spiritual Civilization Scholarship**
2023–2024: **Social Practice Scholarship**